Year 12 Further Maths - Counting principles

Similar questions - Arrangements, selections and adjacency

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1

(a) A committee of 8 people is chosen at random from 7 men and 9 women. Find the probability that the committee contains no more than 3 women. **[4]**

(b) A group of n people, including Mr and Mrs Fermat, are arranged at random in a line. The probability that Mr and Mrs Fermat are next to each other is less than 0.05. Find the smallest possible value of n. **[4]**

2 The 12 members of a team, 5 men and 7 women, stand in a random order in a line for a photograph.

(a) Find the probability that all the men are next to each other. **[3]**

(b) Find the probability that no two men are next to each other. **[4]**

3 Maya claims that she can read minds. A volunteer arranges the integers 1 to n in an order of the volunteer's own choice and Maya states what order she believes was chosen. They agree that her claim will be accepted if she gets the order completely correct, or if exactly two numbers (not necessarily adjacent) are the wrong way round and everything else is correct. They use a value of n such that, if Maya simply guesses the order at random, the probability that her claim will be accepted is less than 0.1%. Determine the smallest possible value of n. **[7]**

4 A club has 18 members, 9 Town members and 9 Country members.

(a) All 18 members are arranged randomly in a straight line. Find the probability that the Town members and the Country members alternate. **[4]**

(b) Nine members of the club are chosen at random. Show that the probability that the number of Town members chosen is no more than r, where $0 \leqslant r \leqslant 9$, is given by $\frac{1}{N}\sum_{i=0}^{r}\left({}^{9}C_{i}\right)^{2}$, where N is an integer to be determined. **[4]**

5 A teacher has 9 different mathematics books: 4 on Algebra, 3 on Calculus and 2 on Trigonometry.

(a) The teacher chooses 4 of the books at random. Find the probability that exactly 2 are on Algebra. **[3]**

(b) The teacher arranges all 9 books in random order on a shelf. Find the probability that the Calculus books are next to each other and the Trigonometry books are next to each other. **[3]**

(c) **In this part you must show detailed reasoning.** Find the probability that exactly 2 of the Calculus books are next to each other but the third Calculus book is separated from those 2 by at least one other book. **[4]**

A music lover has 24 CDs in a random order in a line: 6 are Jazz, 8 are Classical and 10 are Pop.

(d) Find the probability that all 6 Jazz CDs are next to each other. **[3]**

(e) Find the probability that, of the 8 CDs furthest to the left, at least 5 are Jazz. **[4]**

Question 1

(a) Number of women k and men $8-k$: $\dfrac{{}^{9}C_0\,{}^{7}C_8+{}^{9}C_1\,{}^{7}C_7+{}^{9}C_2\,{}^{7}C_6+{}^{9}C_3\,{}^{7}C_5}{{}^{16}C_8}=\dfrac{0+9+252+1764}{12870}=\dfrac{2025}{12870}=0.157$

(b) Treat the couple as a block: $2(n-1)!$ arrangements out of $n!$, so the probability is $\dfrac{2}{n}.\ \dfrac{2}{n}<0.05$ gives $n>40$, so $n=41$

Question 2

(a) Men as a block: $8!\times 5!$ out of $12!$, so $\dfrac{8!5!}{12!}=\dfrac{1}{99}=0.0101$

(b) Arrange the 7 women, then choose 5 of the 8 gaps: the probability is $\dfrac{{}^{8}C_5}{{}^{12}C_5}=\dfrac{56}{792}=\dfrac{7}{99}=0.0707$

Question 3

The number of acceptable orders is $1+{}^{n}C_2$ (the correct order, or any one of the ${}^{n}C_2$ choices of a pair swapped). Probability $=\dfrac{1+{}^{n}C_2}{n!}$. $n=7$: $\dfrac{22}{5040}=0.00437>0.001$. $n=8$: $\dfrac{29}{40320}=0.000719<0.001$. So $n=8$

Question 4

(a) Two patterns (starting with Town or with Country): $\dfrac{2\times 9!\times 9!}{18!}=\dfrac{2}{{}^{18}C_9}=\dfrac{2}{48620}=\dfrac{1}{24310}=4.11\times 10^{-5}$

(b) With i Town members chosen, $9-i$ Country members: ${}^{9}C_i\,{}^{9}C_{9-i}=\left({}^{9}C_i\right)^2$. The total number of selections is ${}^{18}C_9=48620$, so $N=48620$

Question 5

(a) $\dfrac{{}^{4}C_2\,{}^{5}C_2}{{}^{9}C_4}=\dfrac{60}{126}=\dfrac{10}{21}$

(b) Calculus block 3! and Trigonometry block 2!, with the 4 Algebra books: 6 units so $6!\times 3!\times 2!$ arrangements. $\dfrac{6!3!2!}{9!}=\dfrac{1}{42}$

(c) Choose positions for the 3 Calculus books among 9: ${}^{9}C_3=84$. None adjacent: ${}^{7}C_3=35$. All three consecutive: 7. Exactly one adjacent pair and the other separate: $84-35-7=42$. Probability $=\dfrac{42}{84}=\dfrac{1}{2}$

(d) $\dfrac{19!6!}{24!}=1.41\times 10^{-4}$

(e) $\dfrac{{}^{6}C_5\,{}^{18}C_3+{}^{6}C_6\,{}^{18}C_2}{{}^{24}C_8}=\dfrac{6(816)+153}{735471}=0.00686$